

Effects of font size and font style of Traditional Chinese characters on readability on smartphones

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ABSTRACT

This study investigated the effects of Traditional-Chinese character sizes and typefaces on both reading time and subjective preference when participants used 5.5-inch smartphones without limiting their viewing distance. An experiment of 3×3 completely randomized design with eighteen replicates was employed. One hundred and sixty-two participants performed reading comprehension tasks and subjective preference ratings which were used to measure the readability of nine design combinations. The results suggested that the smaller character size was better than the larger one for reading tasks. Additionally, the most favorite typeface was the Kai or Hei style. These findings might be more appropriate than those of previous studies to be generalized for Traditional-Chinese typographic design on smartphones because the real conditions of operating smartphones were considered in this study.

1. Introduction

Font size and typeface are two typographical variables frequently discussed in previous studies of alphanumeric text readings. Those studies concluded that large alphanumeric characters with high legibility would facilitate text readability (Bernard et al., 2003; Dyson and Kipping, 1998; Garcia and Caldera, 1996; Kolers et al., 1981; Paterson and Tinker, 1947). However, they did not present consistent findings on the effects of alphanumeric typefaces on readability. Some indicated that serif characters were more legible than sans serif because serifs could offer hints for character recognition (Arditi and Cho, 2005; Banerjee et al., 2011; Dogusoy et al., 2016). Others found serif fonts offered no reading benefits because the strokes of serif fonts were of complexity which might serve as noise and jumbled characters (Moret-Tatay and Perea, 2011; Ali et al., 2013).

In English, an alphabetic writing system, the words are separated by spaces; however, written Chinese text is formed by strings of equally spaced symbols not composed of letters, nor separated by spaces. In English reading, the spaces between words help to demarcate the word boundary (Rayner et al., 1998). However, Chinese readers have to depend on lexical knowledge to segment Chinese characters into words (Li et al., 2009). Besides, Chinese characters have more structural details and are visually more complex in form than alphanumeric characters. Possibly, suggestions regarding readability of the font size and typeface of alphanumeric text could not be generalized to Chinese text because of the differences between alphanumeric and ideographic writing systems. Therefore, it is necessary to explore the effects of font size and

typeface of traditional-Chinese characters on reading performance.

1.1. Font size

Previous studies concluded that large Chinese characters had high legibility (Dobres et al., 2017, 2018) and that the legibility of Chinese characters significantly influenced readability (Chi et al., 2003; Chan and Lee, 2005; Huang et al., 2009; Chan and Ng, 2012; Liu et al., 2016). They found that large Chinese characters were superior to small ones in terms of reading comprehension, reading speed, preference, and fatigue. For example, Chan and Lee (2005) suggested that the 14 pt (point) Chinese character (38' arc) was superior to the 10 pt one (27' arc) in terms of performance on total reading time, reading comprehension, and subjective preference in a 15-inch color CRT. (The point (pt), a size unit used to specify the height of a character, is equivalent to 1/72 of an inch.) They also found that subjects reported less fatigue, more reading ease, and more comfort when reading text with larger font sizes. Moreover, Huang et al. (2009) found that the reading speed of a large Chinese font (ex. 26' arc) was significantly faster than a small font (ex. 14' arc) when the viewing distance of the mobile phone was fixed. In addition, Chan and Ng (2012) found that 14-point characters were rated significantly superior to 12-point characters in a 17-inch LCD monitor.

1.1.1. Reading behaviors on handheld devices

Nevertheless, the conclusions reached in previous studies were based on the assumptions that the viewing distance to the screen was

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confined to a fixed distance and that the experiment must be performed under well-controlled conditions to comply with the ergonomic principles of desktop displays (ex. OSHA 3092, 1997). However, everyday smartphone reading behavior bears little resemblance to such experimental conditions. By bending his elbow, the smartphone user can hold the phone and adjust his viewing distance at will, which puts the screen much closer than the suggested ergonomic VDT design for comfortable reading. Besides, to see the text clearly, users can adjust their viewing distance to attain an optimal visual angle subtended to the retina: short (near) for small character sizes; and long (far) for large character sizes. Consequently, both large and small characters may be legible enough for readers.

Hence, this study predicts that font size may not significantly affect reading performance on smartphones because the visual angles of various character sizes formed on the retina may be similar with adjustment to the viewing distance. The findings of Darroch et al. (2005) supported this assertion. They arranged a proof-reading experiment, which did not set up a rigid viewing distance, and asked subjects to hold a PDA handheld device and to read the messages written in English from a comfortable position. Provided that the English text messages were all legible, they found no significant difference in text-reading performance.

1.1.2. Scrolling effect

However, due to the size limit of smartphone displays, the text on the small screen may overflow onto multiple screens, prompting users to scroll the screen in order to read the text. Previous studies suggested that scrolling might negatively affect alphanumeric reading performance (Morrison and Duncan, 1988; Piolat et al., 1997), including reduction of reading comprehension on full-size displays (Sanchez and Wiley, 2009), and memory overload when reading on small screens (Sanchez and Goolsbee, 2010).

Text written with a large font may cost more space and lengthen the text on the screen, and readers may increase the frequency of scrolling to read through the lengthened text. Consequently, text written with large characters may degrade reading performance owing to increased scrolling behavior, and text with a small font may facilitate better reading performance when the characters are legible enough on smartphones. Therefore, this study questions whether a large font has better readability than text in a small font, and hypothesizes that the readability of text in small characters on smartphones is superior to that in larger ones.

1.2. Typefaces

1.2.1. Complexity of typefaces

Among Chinese typefaces, Kai, Ming, and Hei styles are used in computer-display text extensively (Cai et al., 2001). A Kai typeface that resembles handwritten calligraphy is one kind of serif characters (Fig. 1a); the stroke width from start to end is not consistent, and the stroke extremities have curved shapes. A Ming typeface (Fig. 1b), which has simpler stroke patterns for printing, is similar to Western serif fonts such as Norman. Unlike Ming typefaces with serifs, a Hei typeface (Fig. 1c), which has a consistent stroke width with minimalistic design

aesthetics, is similar to Western sans-serif typefaces such as Verdana. Among the Chinese characters, the Kai typeface is the most complicated, the Ming typeface is the less complicated, and the Hei typeface is the least complicated (Chi et al., 2003).

1.2.2. Character size and its actual size

Each Chinese character on the display screen is confined within a square frame which is called the Character Frame, as shown in Fig. 1. Character size is measured by the size of the character frame (Tung et al., 2017). For example, a character size is 12 points when the length of its character frame is 12 points. Aligned with the vertical and horizontal center of the character frame, there is a smaller box called the Letter Face, which contains the actual glyph (Tung et al., 2017). The black space between the letter face and the character frame shows that the actual size of a character is less than its frame size. Besides this, the sizes of letter faces are different for varied typefaces with identical character frame. In other words, the actual sizes of Chinese characters of Hei, Ming, and Kai styles are not equal although they have an identical frame size. The typefaces arranged in order of their actual sizes from large to small are Hei, Ming, and Kai (Chi et al., 2003).

1.2.2.1. Complexity and legibility. Previous studies indicated that the simpler Chinese typefaces were, the greater the legibility (Shieh et al., 1997; Cai et al., 2001; Dobres et al., 2016). Hence, previous studies showed that Characters with Ming typefaces had better legibility than those with Kai (Shieh et al., 1997; Cai et al., 2001), and Hei styles were more legible than Ming and Kai styles (Chi et al., 2003; Dobres et al., 2016).

1.2.2.2. Legibility and reading performance. As mentioned above, the legibility of Chinese characters significantly influences text readability; and previous studies claimed that simple typefaces could promote text readability because of their high legibility (Shen et al., 2009; Lai et al., 2007). For example, Shen et al. (2009) showed that Ming styles had higher search accuracy than Kai styles; and Lai et al. (2007) indicated less reaction time and lower error rates for Hei styles than Kai styles. They all attributed the outcomes to the Hei style being more legible than the Ming or Kai styles because the shape of Hei styles was simpler than the others (Shen et al., 2009; Lai et al., 2007).

We find these conclusions questionable. The order of actual character (glyph) size of these typefaces from large to small (Hei, Ming, and Kai) is the same as their order from simple to complex. A typeface with greater actual size also has greater legibility than one smaller in size when viewing distances are fixed. Consequently, the effect of font complexity is confounded with that of actual size for those studies. Restated, it is questionable whether typefaces with simple shapes are necessarily more legible.

This study argues that the legibility of a font style may be determined by its actual size rather than its shape complexity. The visual angles of the different typefaces subtended to the retina would be large enough to read the typefaces clearly when users are allowed to adjust their viewing distances arbitrarily. Hence, this study hypothesizes that the effect of typefaces on readability may not be significant on a

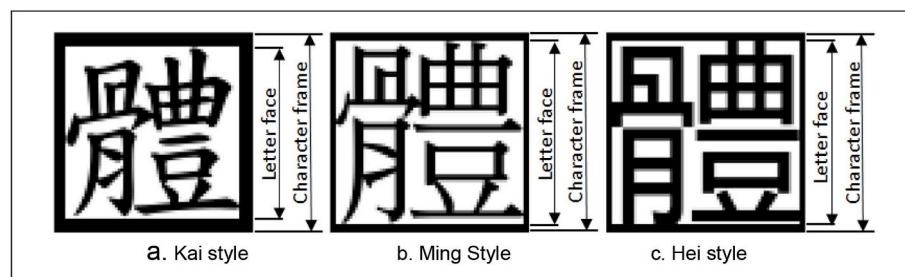


Fig. 1. The font styles of Chinese characters.

smartphone when users can adjust their optimal viewing distance. Restated, this study tried to investigate the effects of Traditional Chinese character sizes and typefaces on text reading time and subjective preferences on smartphones.

2. Methods

The experiment is a completely randomized 3 (character sizes) $\times$ 3 (typefaces) between-subjects design. There are a total of nine design combinations in the experiment. Each design combination is replicated 18 times. The character size has three levels: 10 pt, 12 pt, and 14 pt. The typeface has three levels including Hei, Ming and Kai styles.

2.1. Participants

One hundred and sixty-two undergraduate students from Medical Colleges in Taiwan participated in the experiment. Subject mean age was 20.83 years. (SD = 2.60 years). Each subject performed only one experimental run. These participants had taken the Chinese language portion of the General Scholastic Ability Test (GSAT), which has scores ranging from 0 to 15, and all had scores over 14, indicating high Chinese reading competence. All subjects reported normal ocular health with no clinically significant anomalies of accommodation. Before participating in the test, all subjects consented to perform the trials by signing the agreements outlined by the committee on Governance Framework for Human Research Ethics of National Cheng Kung University in Taiwan.

2.2. Materials

A 5.5-inch smartphone (ZenFone 2 Laser model ZE550KL) with a display resolution of 267 ppi was used to display a reading comprehension test. The smartphone was programmed with Android 5.0 (Lollipop) to control the experimental procedures and collect data.

Four essays were chosen from reading comprehension tests used in an elementary school in Taiwan. The lengths of the four essays, including punctuation marks, were 784, 728, 424 and 801 characters, respectively. Each of the four essays appeared once in each experimental trial.

2.3. Tasks and procedures

First, the subjects were asked to perform reading comprehension tests. The smartphone screen presented a Preparing Page with a blue band showing white traditional-Chinese words meaning “start timing” (Fig. 2). Subjects began the test by tapping on the blue band. Subjects were reminded to take a breath before they began.

After subjects tapped on the blue band, an Essay Page appeared showing the first essay in one of the nine design combinations, and the starting time was recorded by the programmed smartphone. Subjects had to slide the screen to read the essay because its length overflowed the screen size. Fig. 3 shows the first essay with its two successive pages. Four multiple-choice questions were presented at the end of each essay. Subjects were asked to choose the correct answers as quickly as they could.

At the bottom of the Essay Page was a blue band with white Chinese words (“答案確定進入下一則故事”) meaning “finished and go to next”. As soon as possible after completing the multiple-choice questions, subjects tapped on this band, which began the second essay comprehension test, following the same format as the first. Subjects continued in this way to read through the four essays in sequence and answer multiple-choice questions. After completing the multiple-choice questions for the last essay, they immediately again tapped the “finished and go to next” band, and the programmed smartphone recorded the ending time and the correct answer rate.

Next, subjects were shown the text of an essay on the “Standard Typesetting Page” as shown in Fig. 4, and were given an opportunity to



Fig. 2. The Preparing Page which has a blue band with white Chinese words, “開始計時” which means “start timing”. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

express their preference of typeface and size by comparing this with the Essay Pages they had viewed before. On the “Standard Typesetting Page”, the typeface and character size were Hei style and 12 pt, respectively. The “Evaluated Page” (shown in Fig. 5) presented the first essay the subject had just read with the type font and size combination they had seen. Subjects rated their subjective preference for the Evaluated Page compared with Standard Typesetting Page. If necessary, participants could jump back and forth between the Standard Typesetting Page and the Evaluated Page by tapping on the screen.

The bottom of Evaluated Page shows a blue band with white Chinese characters (“進入美感評估頁開始評量(可返回)”) which indicates “enter Preference Rating Page.” After participants tapped on the blue band, the Preference Rating Page appeared and presented a questionnaire including five choices, as shown in Fig. 6. The five choices from top to bottom are “prefer most,” “prefer,” “no opinion,” “less prefer” and “least prefer,” scored as “5,” “4,” “3,” “2” and “1,” respectively. After making their choice, participants tapped on the blue box at the bottom right to complete the trial. The Chinese characters inside the blue box are “評量完成” which means “Evaluation completed.”

Each subject spent less than ten minutes taking the tests. Familiarization was based on one training trial using an additional essay with different typographic variables.

2.4. Performance measures and data analysis

Reading time, a performance measure, was calculated for each part of the test by subtracting the starting time from the ending time. Preference rating scores ranged from 1 for “prefer most” to 5 for “least prefer.” Analysis of Variance was employed to analyze the data. All calculations were made by using SPSS.

3. Results

Table 1 indicates that there was a significant main effect of Character Size on Reading time ($F(2, 153) = 11.083, p < 0.001$), whereas



Fig. 3. The Essay Page which shows the first essay. The first half and second half of the essay is shown on the left and right, respectively. Two pages are needed because the text length is longer than the screen.

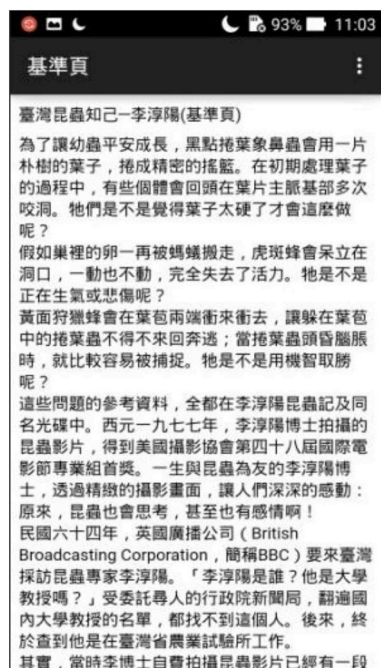


Fig. 4. The Standard-Typesetting Page in the experiment.

the main effect of Typeface was not significant ($F(2, 153) = 0.887$, $p = 0.414$). The effect of Character Size also was not significant ($F(4, 153) = 0.672$, $p = 0.612$).

Table 2 shows the means and standard deviations of reading time for each level of the independent variables. Multiple comparisons using Duncan's method shows that the reading time is the longest when using the 14 pt character (552.19sec), followed by 12 pt (448.02 s) and 10 pt (430.56 s).

Table 3 indicates that the main effect of Typefaces was significant on Preference rating ($F(2, 153) = 6.185$, $p = 0.03$), whereas the main effect of Character Size was not significant ($F(2, 153) = 0.446$,

$p = 0.641$). The interaction of Typefaces and Character Size also was not significant ($F(4, 153) = 0.826$, $p = 0.511$).

Table 4 shows the means and standard deviations of preference ratings for each level of the independent variables. Multiple comparisons using Duncan's method shows that preference ratings for both the Kai style (3.54) and Hei style (3.20) do not significantly differ, but ratings for the Ming style (2.70) are significantly lower. That is, subjects preferred text written in Kai or Hei styles to Ming style significantly.

Table 5 shows no significant effects of either typeface ($F(2, 153) = 0.057$, $p = 0.944$) or Character Size ($F(2, 153) = 0.400$, $p = 0.671$) on correct answer rates. The interaction of Typefaces and Character Size also was not significant ($F(4, 153) = 1.40$, $p = 0.237$).

4. Discussion

4.1. Font size

Previous studies indicated that large fonts promoted better reading performance than small fonts because large fonts provided better legibility on desktop screens. However, the results of this study do not indicate a superiority of large font in reading performance. Conversely, the results show that subjects spent the least amount of time on reading the text written in the smallest font (i.e., 10 pt) and spent the most amount of time on the largest characters (i.e., 14 pt). The combination of two reasons might explain these findings: adjusting viewing distance at will and scrolling on the screen.

First, the effect of character sizes on reading time might be eliminated because subjects could adjust optimal viewing distances to see characters legibly in this study. As previous studies confined the viewing distance to a fixed distance to execute reading tasks on desktop displays, the large characters facilitated the reading performances because of their large visual angle subtended to the retina. However, in this study, the subject could hold the smartphone and adjust his viewing distance at will to attain an optimal visual angle subtended to the retina to see the characters clearly. In this situation, font size might not significantly affect reading performance because the visual angles of different character sizes formed on retina might be similar for various



Fig. 5. The Evaluated Page in the experiment; the picture on the left is the top part of the page; on the right is the bottom of the page.

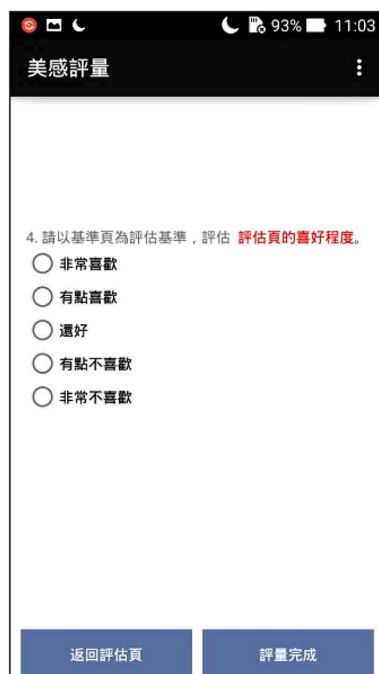


Fig. 6. The preference rating page.

Table 1
Analysis of variance for the reading time (sec).

Source of Variation	Sum of Squares	df	Mean Square	F ₀	P-value
Typeface	31998.725	2	15999.363	.887	.414
Character Size	399802.597	2	199901.298	11.083	.000
Typeface x Character Size	48481.915	4	12120.479	.672	.612
Error	2759498.296	153	18035.937		
Total	3239781.533	161			

Table 2

Means and standard deviations of reading time for each level of independent variables.

	Level	N	Mean	Standard Deviation	Duncan Groups
Character Size	10 pt.	54	430.56 s	139.17	A
	12 pt.	54	488.02 s	131.02	B
	14 pt.	54	552.19 s	130.59	C
	Kai	54	505.42 s	128.67	A
Typeface	Hei	54	493.80 s	160.94	A
	Ming	54	471.55 s	134.42	A

Note: Different letters in Duncan group indicate significant differences at 0.05 levels.

Table 3

Analysis of variance for the preference ratings.

Source of Variation	Sum of Squares	df	Mean Square	F ₀	P-value
Typeface	19.000	2	9.500	6.185	.003
Character Size	1.370	2	.685	.446	.641
Typeface x Character Size	5.074	4	1.269	.826	.511
Error	235.000	153	1.536		
Total	260.444	161			

Table 4

Means and standard deviations of preference ratings for each level of independent variables.

	Level	N	Mean	Standard Deviation	Duncan Groups
Character Size	10 pt.	54	3.07	1.29	A
	12 pt.	54	3.28	1.17	A
	14 pt.	54	3.10	1.36	A
	Kai	54	3.54	1.22	A
Typeface	Hei	54	3.20	1.17	A
	Ming	54	2.70	1.30	B

Note: Different letters in Duncan group indicate significant differences at 0.05 levels.

Table 5
Analysis of variance for correct answer rates.

Source of Variation	Sum of Squares	df.	Mean Square	F ₀	P-value
Typeface	0.111	2	0.056	0.057	0.944
Character Size	0.778	2	0.389	0.400	0.671
Typeface x Character Size	5.444	4	1.361	1.400	0.237
Error	148.778	153	0.972		
Total	155.111	161			

character sizes. Second, a large character may increase reading time owing to the increased frequency of scrolling, since the large characters lengthen the texts. Previous researchers indicated that scrolling the desktop displays would negatively impact reading performance (Piolat et al., 1997; Sanchez and Wiley, 2009; Morrison and Duncan, 1988). In this study, the three essays with 14 pt, 12 pt and 10 pt characters cover about 2.5, 2 and 1.5 pages respectively. Subjects might slide the screen more frequently when reading large characters than when reading small characters. Consequently, the reading time for a 14 pt character is the longest, followed by 12 pt characters and 10 pt characters. This confirms that the increased length of text degrades reading performance and that the negative scrolling effect exists for reading on a smartphone.

In summary, adjusting viewing distances may offset any effect of character size on reading time, while reading texts in large characters requires more sliding behavior, increasing reading time. Consequently, we conclude that small characters provide better reading performance than large ones in reading text on smartphones.

The results also demonstrate that preference ratings do not significantly differ among varying character sizes. The reason might be that, because subjects could adjust their viewing distance, the size of characters subtended on the retina would be the same as one another and all varied-size characters of a specific typeface might be perceived as identical in size. As a result, their preference scores do not significantly differ.

4.2. Font style

Previous studies indicated that the Kai typeface is the most complicated, Ming the less, and Hei the least (Chi et al., 2003). They demonstrated that the less complicated a Chinese typeface is, the better its legibility on desktop displays. Therefore, as the legibility of characters could facilitate text readability, it was concluded that there is a significant effect of font style on readability (Dobres et al., 2016; Lai et al., 2007; Chan and Lee, 2005).

However, our test results, suggesting an absence of significant effect of typeface on reading time, are contrary to all previous studies. The actual character size of each typeface may explain the contradiction. In fact, the actual sizes of the Chinese characters of Hei, Ming, and Kai styles are not equal even when they have identical point size on the computer screens (Chi et al., 2003). The typefaces arranged in order of actual size on the screen from large to small are Hei, Ming, and Kai styles. Chinese characters whose actual size is greater have high legibility (Chi et al., 2003; Liu et al., 2016). Therefore, the typefaces arranged by legibility from high to low are Hei, Ming, and Kai styles. This order is the same as the order of typefaces arranged from simple to complex typefaces. The reason why the typefaces affected reading performance in those previous studies may be not their complexity, but their actual sizes measured directly from the displays. That is, the effects of typeface styles on readability found in previous studies correspond to the sizes of the different typefaces used. In this study, as a fixed viewing distance was not set in order to stimulate real behaviors of using smartphones, subjects could adjust their viewing distance to see the characters clearly. The result implies that the reading time may

be not affected by the complexity of typefaces, but by the actual size of typefaces.

However, the effect of typefaces on preference ratings is significant. A Kai character, which resembles handwritten calligraphy with curved shapes on the stroke extremities, might present decorative design aesthetics. A Hei character, a kind of sans serif typeface with solid strokes of consistent width, may exhibit minimalist modernist aesthetics. A Ming character has serifs at the end of strokes, and vertical strokes are thicker than horizontal strokes. A conservative workhorse style, Ming is descended from traditional printing. The simple beauty of the Ming style is not as good as that of the Hei style, and the stroke vivaciousness of the Ming style is also less than that of the Kai style. Preference ratings are used to assess the overall impression and satisfaction about the typefaces, including usability and aesthetic appeal. The preference ratings for the Kai and Hei styles do not significantly differ from each other, but are significantly higher than the Ming style. Previous studies also showed that Kai is aesthetically more pleasing to readers than Ming (Shieh et al., 1997; Yen et al., 2011). The results seem to indicate the existence of a "what is beautiful is usable" stereotype (Tractinsky et al., 2000). Restated, a typeface with a beautiful appearance may be judged positively in all dimensions, including usability, and its perceived beauty may be more influential in affecting user preferences than conventional readability in typography (Norman, 2004). The outcomes might suggest that the either Kai style or Hei style could be an appropriate default typeface to suit user preference on smartphones.

4.3. Reading comprehension scores vs. reading time

Previous studies found no significant effect of Chinese typographic variables on reading comprehension if the participants' performance was measured by the accuracy of untimed multiple-choice comprehension tests (Wang and Chen, 2000; Chen and Chien, 2005; Chan and Lee, 2005). The reason may be the ease of the reading task chosen for these participants. Their reading materials are collected from junior high schools or elementary schools, and the participants are college students with excellent linguistic competence who can achieve high comprehension scores easily. Therefore, the effect of typographic variables on text readability could have been masked by participants' reading competencies when the reading time is not limited.

Our test results, likewise, indicate that the reading comprehension scores are not a useful indicator to find out the optimal typefaces. The essays used in the test, selected from a primary school, were too easy for all participants, whose Chinese test scores of GSAT were over 14, and whose mean correct answer rate was 98.53% in the comprehension tests of the experiment. Accordingly, the effects of character sizes and typefaces on correct answer rates were not significant, neither was the interaction of typefaces and character size.

Nonetheless, the reading time may be an appropriate indicator for measuring the effect of typeface on readability. For example, Chan and Lee (2005) found the reading time of text with Ming styles was significantly shorter than with Li styles when conducting reading comprehension tests. For this reason, we adopted reading time as a performance measure to evaluate the impact of font size and typefaces. The result shows that the effect of character size is significant when the performance measure is reading time. It implies that reading time may be more sensitive than comprehension scores to the subtle influences of typographic variables. Therefore, this study suggests that reading time is a more suitable measure than comprehension score in testing the effect of font size or typeface.

4.4. Adjustable viewing distances

Most previous studies related to Chinese typography on digital screens fixed viewing distances in the experimental trials. However, users can adjust their viewing distance arbitrarily in the real context of using smartphones. In order to simulate the real context in this

experiment, this study does not set a fixed viewing distance, and subjects were encouraged to adjust their viewing distance at any time. Obviously, the evidence in this study does not support previous studies. As explained above, experimentation incorporating normal smartphone usage leads to conclusions different from those of previous studies.

The results of this study are very encouraging. The findings suggest that small Traditional Chinese characters are superior to large ones for reading tasks and that the most favored typeface is the Kai or Hei style, though the effect of typeface choice is not significant on reading performance.

However, for reading the text of smaller characters, the viewing distance to the screen is reduced, and the shorter distance would likely lead to increased strain of ciliary muscles, resulting in eye fatigue. This seems to be a legitimate reason why the smallest characters might not be the most appropriate choice. However, this study does not cover the tradeoff between character size and eye strain, which will be explored in a future study.

5. Conclusions

This study explores the effects size and font type of Traditional Chinese characters on both reading time and subjective preference on smartphones. The results show that the effect of character size on reading performance is significant, with larger characters leading to a slower reading of longer texts by requiring increased scrolling. Consequently, the small characters are better than the large ones on reading time.

Additionally, the results show that the effect of typeface on preference ratings is significant, and suggest that the preferred typeface is the Kai or Hei style. Finally, owing to consideration of the everyday context of smartphone use, the findings of this study might be considered more appropriate than those of previous studies for generalization to the typographic design of the Traditional Chinese text on smartphones.

Relevance to industry

The findings of this study can be used in determining the optimal typeface and font size of traditional Chinese characters for text display on smartphones.

Conflicts of interest

The author declares no conflict of interest.

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